Cheat Sheet: Derivatives

Notation: $f'(x) = \frac{df}{dx}$ is the slope of f at x. Below, c, n, a are constants and f, g are functions of x.

Basic functions

Rule	General	Example
Constant	$(c)' = 0$	$(7)' = 0$
Power rule (any real n)	$(x^n)' = nx^{n-1}$	$(x^5)' = 5x^4$
roots	$(\sqrt{x})' = (x^{1/2})' = \frac{1}{2\sqrt{x}}$	$(\sqrt[3]{x})' = (x^{1/3})' = \frac{1}{3}x^{-2/3}$
reciprocals	$(\frac{1}{x})' = (x^{-1})' = -\frac{1}{x^2}$	$(\frac{1}{x^3})' = (x^{-3})' = -\frac{3}{x^4}$
Exponential	$(e^x)' = e^x$ $(a^x)' = a^x \ln a$	$(5e^x)' = 5e^x$ $(2^x)' = 2^x \ln 2$
Logarithm ($x > 0$)	$(\ln x)' = \frac{1}{x}$ $(\log_a x)' = \frac{1}{x \ln a}$	$(\ln x^2)' = (2 \ln x)' = \frac{2}{x}$ $(\log_2 x)' = \frac{1}{x \ln 2}$
Sine, cosine	$(\sin x)' = \cos x$ $(\cos x)' = -\sin x$	$(2 \sin x - \cos x)' = 2 \cos x + \sin x$

Combining functions

Rule	General	Example
Constant factor	$(c \cdot f)' = c \cdot f'$	$(3x^2)' = 3 \cdot 2x = 6x$
Sum rule	$(f + g)' = f' + g'$	$(x^3 + \sin x)' = 3x^2 + \cos x$
Product rule	$(f \cdot g)' = f'g + fg'$	$(x^2 e^x)' = 2xe^x + x^2 e^x$
Quotient rule ($g \neq 0$)	$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$	$\left(\frac{e^x}{x}\right)' = \frac{e^x \cdot x - e^x \cdot 1}{x^2} = \frac{(x-1)e^x}{x^2}$
Chain rule "outer′ · inner′"	$(f(g(x)))' = f'(g(x)) \cdot g'(x)$ Leibniz: $\frac{df}{dx} = \frac{df}{dg} \cdot \frac{dg}{dx}$	$\left(e^{-x^2}\right)' = e^{-x^2} \cdot (-2x)$ (outer e^u, inner $u = -x^2$)
nested	apply repeatedly, from the outside in: $\frac{df}{dx} = \frac{df}{dg} \cdot \frac{dg}{dh} \cdot \frac{dh}{dx}$	$\left((\ln(1 + x^2))^3\right)'$ $= 3(\ln(1 + x^2))^2 \cdot \frac{1}{1+x^2} \cdot 2x$

Several variables

Rule	General	Example
Partial derivative	Differentiate w.r.t. one variable, treat all others as constants	$f = x^2 y + 3y$: $\frac{\partial f}{\partial x} = 2xy$, $\frac{\partial f}{\partial y} = x^2 + 3$
Gradient	$\nabla f = \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\right)^T$ points in the direction of steepest ascent	$\nabla f(x, y) = (2xy, x^2 + 3)^T$

Frequently used in deep learning

Rule	General	Example
Sigmoid	$\sigma(x) = \frac{1}{1+e^{-x}}$	$\sigma'(x) = \sigma(x)(1 - \sigma(x))$
tanh	$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$	$\tanh'(x) = 1 - \tanh^2(x)$
ReLU	$\mathrm{ReLU}(x) = \max(0, x)$	$\mathrm{ReLU}'(x) = 1$ for $x > 0$, 0 for $x < 0$ (not differentiable at 0; in practice set to 0)
Squared error	$(y - \hat{y})^2$	$\frac{\partial}{\partial \hat{y}}(y - \hat{y})^2 = -2(y - \hat{y})$